

AI ROI Calculator — Human vs AI + HITL

Should you optimize the human process, or bring in AI with a human-in-the-loop? Every input is a lever. Move a slider and the risk-adjusted P&L, the break-even projection, and the recommended human/AI split all recompute live. Same math as the Python engine (`roi_tracker`).

pricing: editable · auto-filled from litellm when the engine runs

INPUTS

The CFO interview, as sliders. Type an exact number in any value box.

1 · Current human process (baseline)

What the work costs *today*, with people - and how much of it there is.

Human cost / transaction (as-is)

4.2

Fully-burdened operator rate x minutes per task. The number every alternative must beat.

Monthly transaction volume

100000

How many transactions per month. Scales every per-unit number.

2 · Optimize-the-humans alternative

Before buying AI: how much cheaper could the *human* process get with lean / tooling, and what would that cost to do?

Human cost reduction from optimization (%)

20

Realistic efficiency gain from redesigning the manual process.

One-time optimization CapEx

40000

Cost to redesign / retrain / re-tool the human process.

3 · Model pricing & routing (AI path)

Live token prices per 1M tokens. Router sends simple work to the cheapest fallback, complex work to the cheapest flagship.

● = auto-selected

MODEL	INPUT \$/1M	OUTPUT \$/1M
● claude-opus-4-8 flagship	15	75
gpt-5 flagship	10	30

MODEL	INPUT \$/1M	OUTPUT \$/1M
gemini-2.5-pro flagship ●	1.25	10
ollama-open-llm fallback ●	0	0
gpt-5-mini fallback	0.25	2
gemini-2.5-flash fallback	0.3	2.5

4 · Task mix & token size

Complex-task share (routed to flagship) 30

Share of traffic hard enough to need a frontier model.

Avg input tokens / txn 1200

Avg output tokens / txn 400

5 · Infrastructure & build cost (AI path)

Vector DB retrieval / txn 0.0004

Total AI CapEx (build + fine-tuning) 180000

One-time build spend. Drives the break-even projection.

Amortize CapEx over (transactions) 5000000

Maintenance OpEx / month 12000

6 · Risk (the Hallucination Tax)

AI is not free of consequence. These deduct from AI savings.

Value at Risk / catastrophic incident 25000

Damage of one bad hallucination or compliance failure in production.

Incident rate (per 100,000 outputs) 5

HITL review cost / flagged output 8

HITL rate (% of outputs reviewed)

5

7 · Process sensitivity (drives the split)

How error-costly and judgment-heavy is this work? High sensitivity keeps more of it human, even when AI is cheaper.

Process sensitivity

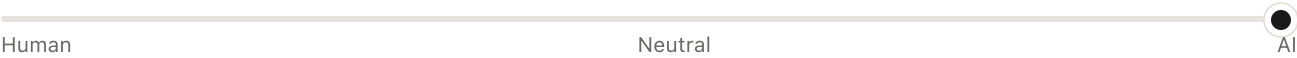
35

At current cost, keep ~35% most-sensitive work human; automate the rest.

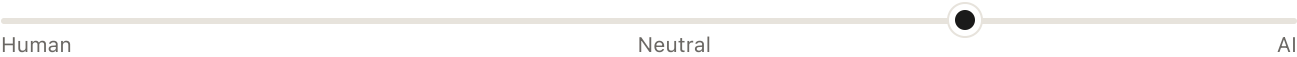
8 · Qualitative measures (Human ↔ AI)

Beyond cost. Slide toward whoever is stronger on each dimension: left = Human wins, right = AI wins.

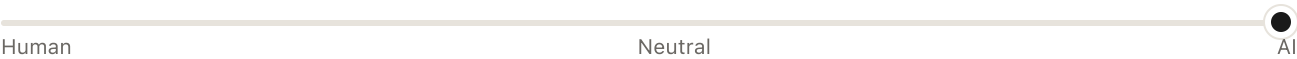
Speed & throughput latency, jobs/hour



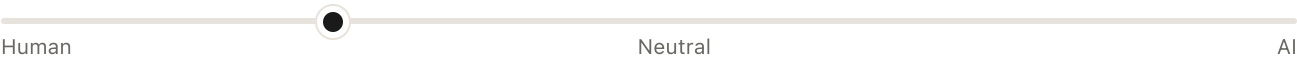
Consistency low output variance



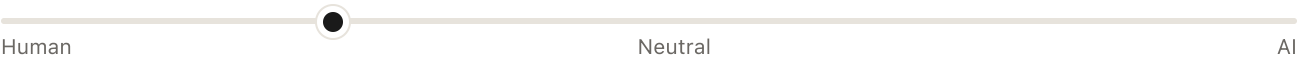
Scalability / surge handle 10x overnight



Auditability explain every decision



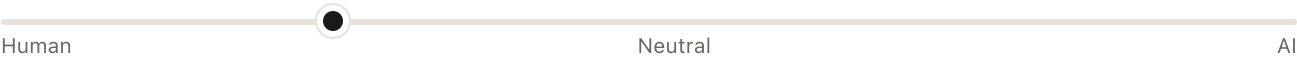
Compliance control regulatory defensibility



Customer experience empathy, nuance



Adaptability to edge cases novel, ambiguous inputs



RISK-ADJUSTED P&L

Live. Recomputes on every change.

NET SAVING / MONTH — AI+HITL VS AS-IS HUMAN

\$239.2K

\$2.87M/yr after risk tax

NET ROI

1,513%

per \$ of AI run cost

CAPEX PAYBACK

< 1 mo

go-AI break-even

AI + HITL COST

\$1.8081

vs \$4.2000 human

RECOMMENDED BLENDED

\$2.3540

\$184.6K/mo saved

HUMAN VS AI + HITL — COST PER TRANSACTION

Three ways to run the same process. Lowest bar wins.

Human (as-is)

do nothing



2.32x

Human (optimized)

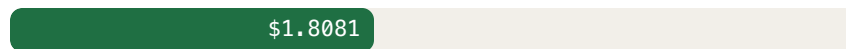
lean + tooling



1.86x

AI + HITL

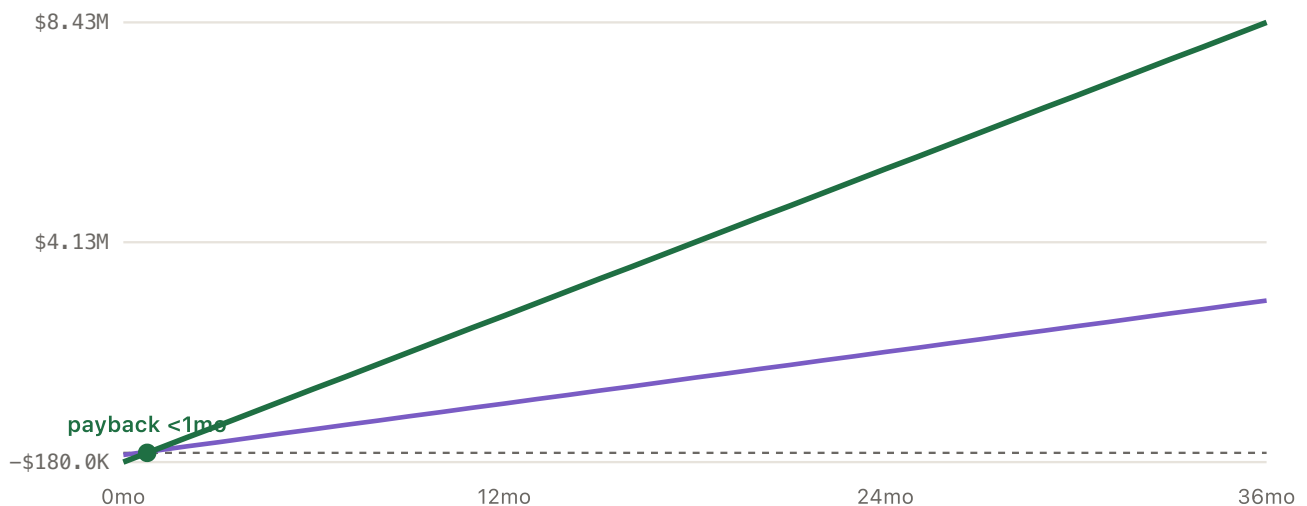
automate + review



cheapest

BREAK-EVEN PROJECTION

Cumulative cash vs staying all-human. Where a line crosses \$0, that investment has paid back.



RECOMMENDED APPROACH

Where to draw the human / AI line, given cost and sensitivity.

Human 35%

AI + HITL 65%

Human up to the most-sensitive 35% of work; AI + HITL for the remaining 65%. Blended cost \$2.3540/txn → **\$184.6K/mo** saved vs doing nothing. AI CapEx pays back in < 1 mo.

Process sensitivity is 35%. Raise it and more work stays human; lower it and automation expands.

QUALITATIVE VERDICT

Cost is one axis. These are the rest.

Speed & throughput

latency, jobs/hour

Consistency

low output variance

Scalability / surge

handle 10x overnight

Auditability

explain every decision

Compliance control

regulatory defensibility

Customer experience

empathy, nuance

Adaptability to edge

cases

novel, ambiguous inputs

Composite AI-favorability

57 / 100

Balanced. Neither side dominates qualitatively (57/100). Let cost and the 35% sensitivity split decide - see the recommendation above.

EXECUTIVE REPORT

One-page executive brief · 100,000 txns/mo · 30% complex → `gemini-2.5-pro` , rest → `ollama-open-`

EXECUTIVE SUMMARY

- Net ROI: **1,513%** per \$ of AI run cost.
- AI CapEx payback: < 1 mo.
- Recommended saving: **\$184.6K/mo** vs staying all-human.

HUMAN VS AI + HITL (COST / TXN)

- Human as-is: \$4.2000 · Human optimized: \$3.3680 · AI + HITL: \$1.8081.
- Cheapest model: **AI + HITL**.

UNIT ECONOMICS (CPO)

- Fully burdened AI cost: \$0.1580/txn (API + infra), before risk.
- Hallucination Tax: **-\$1.6500/txn**.

RECOMMENDED APPROACH

Adopt a 35/65 human/AI split. Keep the most-sensitive work human, automate the rest with HITL. Grow volume past break-even to amortize CapEx faster; wire real telemetry to replace assumed risk rates with measured ones.

[Copy executive report \(Markdown\)](#)[Reset to baseline](#)